

PRD — Category Spark

Product: Blinkit · Growth · Category Exploration

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Status: Prototype deployed, not experimented. No lift figure exists and none is claimed.

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Live prototype	https://ayushbisht-source.github.io/Blinkit/
Review analysis workflow	https://ayushbisht-source.github.io/Blinkit/workflow/
Repository	https://github.com/ayushbisht-source/Blinkit

A note on what this document is. It was written *after* the prototype shipped, from the research, the spec and the code that exist. It is a consolidation, not a forward plan, and where the build corrected an earlier assumption the correction is recorded here rather than smoothed over. §14 lists those corrections explicitly.

1. Summary

Blinkit wants a higher share of Monthly Active Customers buying from at least one **new** category each month. The obvious reading — that users need more encouragement — is wrong: **39 of 40 surveyed have wanted to try something new and didn't**. Desire is not the missing input.

The binding constraint is **evaluation speed**. The app shows an unfamiliar category exactly what it shows a familiar one: a price. For a category you already buy that is sufficient, because you hold the reference point in your head. For one you have never bought, a price with no reference point is uninterpretable — so evaluating it costs time a two-minute fetch session has not budgeted, and skipping is the rational move.

Category Spark supplies the missing reference point at cart review: up to five suggestions, each opening a *different* never-bought category, each carrying a reason drawn from the shopper's own history, a **price anchor** expressed as ₹/week against their own comparable spend, and a trust signal. It shows nothing at all when the data cannot substantiate a reason.

2. Strategic context

The company goal, verbatim from the brief:

Increase the percentage of Monthly Active Customers who purchase products from at least one new category every month. *Examples: a user who buys groceries starts buying pet supplies; a user who buys snacks starts buying personal care; a user who buys household essentials starts buying baby products.*

Formalised as **CER (Category Expansion Rate)** in `docs/00-foundation.md`:

$$\text{CER}(\text{month } M) = \frac{\text{distinct users who purchased from category } C \text{ in } M, \text{ where } C \notin \text{any category that user purchased in } M-1 \dots M-6}{\text{distinct users with } \geq 1 \text{ order in } M}$$

The six-month lookback is load-bearing and shapes the entire product. Without it, a user rebuying mosquito repellent each summer registers as exploring. With it, a repeat purchase in a category tried last month scores **zero** — which is why \$7 ranks first crossovers ahead of repeat prompts.

Why the company should care beyond GMV. The revenue from a ₹99 pet-food pack is negligible. Category breadth is a retention property: a user buying from one category is trivially substitutable — a competitor needs to win one habit. A user buying from five has five habits to dislodge and a basket that is inconvenient to rebuild elsewhere. **This is a moat investment, not a GMV tactic**, and it should be argued as such.

3. Problem

3.1 Evidence base

Instrument	n	What it is
Public review corpus	5,898 collected · 3,372 analysed	Play Store + App Store, 11 apps. 971 judged relevant, 937 extracted, 25 themes
Screeener-matched survey	40	Real respondents, collected 28-29 Jul 2026
Depth interviews	5	Real, async over text, anonymised P01-P05 across three platforms

No interview was fabricated. AI-synthesised personas were generated, tested, and **excluded from the evidence base** — of five hypotheses they produced, one was testable and it was wrong (`docs/02 §7`).

3.2 Root cause

Five whys, from `docs/03-problem-definition.md`:

1. **Why don't they buy new categories?** They arrive already decided — 38/40 "always" or "mostly" know before opening the app. Two people browse.

2. **Why pre-decided?** The app is a *fetch tool* under time pressure. Fast delivery is why 25/40 chose it. The session has a job and a ~2-minute budget.
3. **Why does fetch mode block a new category?** Trying something unfamiliar requires an *evaluation* — fit, price, trust — and that isn't in the budget. 13/40 named comparison time directly.
4. **Why does evaluation take so long?** The information needed isn't present. Users must supply it themselves.
5. **Why isn't it present?** The interface treats an unfamiliar category exactly like a familiar one.

Root cause. The app presents unfamiliar categories with the same information as familiar ones. For a familiar category that is enough, because the user supplies the missing context from memory. For an unfamiliar one it is not — so evaluating it costs time the user has not budgeted, and the rational move inside a two-minute fetch session is to skip it.

Every skip is individually correct. The aggregate outcome is a user who has bought the same things for two years.

In Fogg terms ($B = \text{Motivation} \times \text{Ability} \times \text{Prompt}$): motivation is present (39/40), prompt is present, **ability is the binding constraint** — not affordability, but *can they decide fast enough*. This dictates what will not work: discounts and "explore now" banners push on variables that are already non-zero.

3.3 Existing workarounds

Workarounds are the strongest evidence a need is real — people only build them for problems they have.

Workaround	Evidence	What it reveals
Buy the category elsewhere	D-Mart 8 · Kirana 6 · Amazon 3	The demand isn't missing, it's leaking. Displaced revenue
Wait for a price signal	"cheaper elsewhere" 12 · "too expensive" 8	Price is a proxy for <i>risk</i> . Cheap enough = safe enough to test
Validate outside the app	14 want reviews first	The evaluation happens — on Instagram or in a shop. So does the purchase
Defer indefinitely	"forgot once cart-filling" 3 · distracted 6	Intent evaporates on contact with the reorder flow

3.4 Where the model stops

Two of five interviewees have barriers **no information closes**:

- **P04** — an incumbent kirana used *"like from ages"*, unblocked only by moving house or the shop closing.
- **P05** — a settled belief after one bad delivery: *"I will never buy from blinkit my trust issues for vegetables is still there."*

A third pattern, **category irrelevance** (3 of 5), has no latent demand at all. The addressable population is therefore **smaller than 39/40 implies** — that number counts everyone who ever wanted to try something, not everyone whose barrier a faster evaluation removes. This bounds the sizing in §5, not the design: the agent already declines to show a card in all three cases.

4. Target user

Habitual Narrow Repeaters.

Criterion	Threshold	Support
Order frequency	≥ 3 orders/month	27 of 40
Tenure	≥ 6 months	majority
Category concentration	≤ 2 categories carry most spend	the modal pattern

Why this segment. Already acquired, already transacting, already trusting the platform with money several times a month. Whatever stops them is *behavioural* — the only kind of blocker a software feature can plausibly move inside one product cycle. A segment whose barrier is "has never heard of Blinkit" cannot be fixed by a card in the cart.

Explicitly not the target: users who already cross categories monthly. They are already counted in the numerator; there is no lift to win there.

5. Goals and non-goals

Goals

1. **Raise CER** among Habitual Narrow Repeaters — the share buying from ≥ 1 new category per month.
2. **Reduce evaluation cost** at the point of consideration, without adding session time.
3. **Never show an unsubstantiated suggestion.** Silence is a first-class outcome.

Non-goals

Not doing	Why
A discovery/browse tab	38/40 arrive knowing what they want. There is no audience
An "explore new categories" banner	39/40 already want to. Desire isn't missing
Push notification campaigns	Intent already arrives in-app (Blinkit 16 vs Amazon 3). Nothing to recapture
Discounts on new categories	Moves first trial, already at 50%. Doesn't move monthly recurrence

Not doing	Why
Changing supply, pricing or logistics	Out of scope; this is an information-flow intervention only

6. Success metrics

Primary

CER, treatment vs holdout, quoted always with its six-month lookback.

Secondary

Metric	Why
New-category trial rate	Mode A's direct output; expected to move first
30-day repeat within a newly-tried category	The honest success criterion. Trial without repeat is a discount, not exploration
Distinct categories per user per month	Slower proxy for breadth-as-moat
Card → add-to-cart conversion	Whether the <i>card</i> works, separable from whether the <i>idea</i> works

Guardrails — pre-registered, with kill thresholds

Guardrail	Kill threshold
AOV	any sustained decline
Core-category order frequency	>3% drop
Refund / return rate	any material rise
Card dismissal rate	>60%
Session length at cart review	any material rise

If any threshold trips, the feature is removed regardless of CER movement. Stated in advance because a metric that only ever justifies shipping is not a measurement plan.

7. Solution

7.1 Where it fires

Cart review — after the last item is added, before checkout. The only natural pause in a fetch-mode session: the task is done, attention is free for a few seconds, and the cart is the

richest available signal about who this shopper is today. Firing earlier competes with the task; firing at checkout competes with payment.

7.2 Two modes

Mode	Trigger	Serves
A — First crossover	Never bought from category C	CER (primary)
B — Second purchase	Bought C for the first time 14–45 days ago, hasn't returned	Secondary metric (30-day repeat)

Mode A leads; Mode B takes the last slot. A Mode B repeat fires inside the six-month lookback by construction, so it contributes **zero to CER** however well it converts. Mode B stays in the product because it defends the primary metric's *meaning* — it is what stops CER being satisfied by one-off trials that never come back — but it is not itself a CER event.

Measured against the seeded shoppers: **35 of 36 row entries would register in CER**, and the single one that would not is the Mode B card (`mvp/evals/cer-audit.mjs`).

7.3 The row

Up to **five** suggestions, each opening a **different** never-bought category.

Deduplicating by *category* rather than by SKU is the entire point: CER counts customers buying from ≥ 1 new category, so five products from one category is five shots at one target while five categories is five targets.

The trade this makes. The spec originally argued for exactly one option, because comparison time is the #2 blocker (13/40) and one option leaves nothing to compare. Five reopens that. The blocker is now **not closed by the layout**; it is mitigated only by each entry carrying its own price anchor, so the comparison is pre-computed per option. Whether that mitigation suffices is an open empirical question (§13).

Rows are routinely shorter than five — mean 3.6 across ten seeded shoppers — because a candidate with no substantiated reason is dropped rather than padded.

8. Functional requirements

Hard rules — each asserted in the eval suite, not just in prose

#	Rule	Enforcement
R1	One impression per session; max one per 72 hours	Cadence rule; the row is one impression, not five
R2	Never a category the user already buys (Mode A)	Eval 1 + eval 8

#	Rule	Enforcement
R3	Only in-stock SKUs from the live catalogue; fail closed to no card	Eval 2 + eval 8
R4	The reason must be true — no substantiated reason means no card	Eval 3
R5	Never a category the user has closed (incumbent supplier, settled distrust)	Eval 7 + eval 10
R6	Every entry in the row opens a distinct new category	Eval 9
R7	The row leads with a CER-eligible suggestion	Eval 12
R8	No new screen; no navigation	Design constraint

Every check is mutation-tested — a rule whose check cannot fail is not evidence. Current state: **11 pass, 1 reports NOT RUN** (no seeded shopper has thin enough history to exercise it).

Card anatomy — each line closes a named blocker

Element	Blocker closed	Survey n
Reason from the shopper's own history	"not sure it'll suit my need"	14/40
Price anchor — ₹/week vs <i>their own</i> comparable spend	"priced higher than usual" / "price feels risky"	15/40
Trust signal from similar buyers	"don't trust the brand" / "want reviews first"	26/40
Smallest available pack	"too expensive" / "found cheaper elsewhere"	20/40

The price anchor is the non-obvious part. Showing ₹99 is a price. Showing "*~₹65/week · your usual runs ~₹68/week*" is an anchor — it answers "is this fair?" using the user's own spending as the reference, which is the comparison they said they don't have time to do manually. **Doing that computation for them is the feature.**

Suppression cases — showing nothing is a designed outcome

- Thin history (<2 orders) → no card
- No never-bought category with a substantiated driver → no card
- Category closed by the user → no card, reason logged as `category_closed_by_user:<category>`
- No in-stock trial pack → no card

9. User flow

1	Fetch-mode session	shopper opens the app knowing what they want (38/40)
2	Basket fills	their usual list – the agent does not interrupt the task
3	Cart review	the only natural pause: task done, payment not started
4	Row appears	up to 5, each opening a different never-bought category
5	Add, or dismiss	dismissal is logged as a guardrail, not hidden

10. Technical design

10.1 Pipeline — decisions deterministic, model confined to wording

Step	What it does	Model?
1 Eligibility	Mode A / Mode B / neither	No — rules, fails closed
2 Profile	Order history → owned categories, cadence, declared signals	No
3 Candidates	Never-bought categories, minus any closed	No — and this is the only list an LLM may choose from
4 Price anchor	₹/week for candidate vs their comparable spend	No — division
5 Compose	One reason line + one trust line	Yes — the only step

Mode, category, product and anchor are identical with or without an LLM; only the wording differs. Copy is pre-generated in CI where the key lives, and discarded at render if the agent picked differently than it did at build time — a stored line about a different product would be a false statement about this shopper.

A demo that a quota reset can break is not a demo.

10.2 Ranking

Signal	Weight	Rationale
Declared lifestyle signal (pet owner, parent, skincare)	100	A fact about <i>this person</i>
Category adjacency	10	A fact about a <i>population</i>
Cheaper trial pack	0–5	Lowers the risk 20/40 named

The brief's three named crossovers are reached by declared signals, not by the adjacency graph, and eval 12 asserts this **on rank, not membership** — with five slots the goal category still appears even when signal scoring is gutted.

10.3 Known weakness

Measured on the seeded shoppers: **7 of 36 reasons rest on a declared fact about the person; 29 rest on a population pattern.** The row made this worse — with one card the lead pick was usually the declared signal; scaling to five filled the extra slots with adjacency, the weakest evidence available. Copy was rewritten to stop the weak case impersonating the strong one, but that is mitigation, not a fix. **A shopper who declares no signal has nothing but adjacency.** This is the ceiling without a richer profile, and it is the next thing to fix — *better signals, not more slots.*

11. Instrumentation

```
nudge_shown           { user_id, mode: A|B, category, product_id, anchor_favourable, ts }
nudge_dismissed       { user_id, mode, category, ts }
nudge_add_to_cart     { user_id, mode, category, product_id, ts }
new_category_purchased { user_id, category, order_id, first_ever_in_category: bool, ts }
category_repeat       { user_id, category, days_since_first, ts }
nudge_suppressed      { user_id, reason, ts }
```

Two of these carry more weight than they look:

- nudge_suppressed** — the agent declines whenever it can't substantiate a reason. Logging *why* tells you whether suppression is working as designed or silently swallowing the audience. **A feature that never fires can't fail, and can't succeed either.**
- anchor_favourable** — the price anchor is the central claim. If cards convert only when the anchor is favourable, the anchor is doing the work. If conversion is flat across both, it isn't — and the core hypothesis is falsified.

12. Rollout and experiment

Design	90/10 holdout, randomised at user level
Unit	User — CER is a per-user monthly measure
Duration	Minimum 4 weeks
Segment	Habitual Narrow Repeaters
Primary comparison	CER, treatment vs holdout

User-level, not session-level: the metric asks whether a *person* bought a new category this month. Session randomisation would contaminate both arms.

Four weeks minimum: Mode B fires 14–45 days after a first crossover. A two-week test measures Mode A only.

Analysis fixed in advance: one primary metric, no fishing across secondaries; guardrails evaluated *first*; no mid-flight stopping on a favourable early read; segment cuts exploratory only.

13. Risks and open questions

Risk	Mitigation / status
Five options reintroduce the #2 blocker (comparison time)	Each entry carries its own pre-computed anchor. Unresolved empirically — session length at cart review is a kill guardrail
Most reasons rest on population patterns, which the research says users ignore	Copy no longer impersonates personalisation. Real fix needs richer declared signals
Trial rises, repeat doesn't	Named as a falsifier. Would mean the feature manufactures one-offs
Addressable population smaller than 39/40 implies	Sizing expressed as a formula over segment share, not a headline number
Card reads as noise in a 2-minute session	Dismissal >60% is a kill threshold

Open questions

1. **Does the price anchor actually do the work?** Testable via `anchor_favourable` in the first run.
2. **Is five the right number?** Chosen for metric coverage, not tested against one or three.
3. **Can declared signals be obtained at scale** without an intrusive onboarding survey?
4. **Does suppression cover too much?** `nudge_suppressed` volume answers this on day one.

14. Corrections made during the build

Recorded rather than smoothed over, because each changed the product.

Correction	What was wrong	What changed
Mode B priority	The spec called Mode B "the metric-moving mode". A repeat 14–45 days after first trial sits <i>inside</i> CER's six-month lookback and scores zero	Mode A now leads; Mode B takes the last slot and is attributed to the secondary metric
Validation reporting	Two checks printed PASS over zero observations — grounding measured on an empty set, hold-out compared training data to itself	Both rewritten; checks that can't run now report NOT RUN
Generic copy	The adjacency reason read "most people who do also keep X stocked" — a population claim worn as personalisation, which F3 says is exactly what fails	Rewritten to own being a pattern and end on a checkable fact

Correction	What was wrong	What changed
Degenerate clustering	Clustering collapsed to one theme at 100% prevalence and reported success	Merge threshold recalibrated on homogeneity; non-degeneracy asserted in code

15. Out of scope / future

1. **Process the remaining corpus.** 2,526 documents are collected but not gated or extracted; enrichment is resumable, so this is a re-run, not a rebuild.
2. **Reddit and forum sources.** The collector is implemented and the authenticated path written; a CI run returned HTTP 403 on 117 of 117 requests (Reddit blocks datacenter IPs for anonymous reads). Needs credentials only.
3. **Richer declared signals** — the single highest-leverage improvement, per §10.3.
4. **Real order histories.** The prototype runs on synthetic data: it demonstrates the mechanism, not real-world lift.
5. **Hand-labelled gold set** (~50 documents) to turn the Cohen's κ check from NOT RUN into a number.

16. Appendix — what is and isn't established

Established: the barrier ranking (two independently collected datasets agree on the same top three); that crossover is event-triggered and non-repeating; that a review corpus structurally cannot answer questions about behaviour *before* or *across* purchases (Q3, Q4, Q7 score 2.9%–14.8% coverage while Q2, Q5, Q6, Q8 score 57%–76%).

Not established: any lift figure; any "X% of users" claim from $n=5$; whether the price anchor is the active ingredient; whether five options is better than one.

The honest summary: this project has a falsifiable hypothesis, a defined primary metric, kill criteria fixed before results exist, and a prototype whose every decision is asserted by tests. It does not have evidence that the feature works, and does not claim any.